

Market and Patent Research Report

Topic: Autonomous Robotic Systems for the Removal of Invasive Plant Species

Case Study: Giant Hogweed and Similar Neophytes

Course: Product Development – From Need to Market

Institution: Hochschule Kaiserslautern

Semester: SS 2025

1. Introduction

Invasive alien plant species such as *Heracleum mantegazzianum*, *Lupinus polyphyllus*, and *Jacobaea vulgaris* pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing.



Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2. Market Overview and Technological Landscape

2.1. RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum*.
- **Advantages:**
 - Non-chemical and environmentally safe.
 - Minimal soil disturbance.
 - Suitable for robotic integration. [Carbon Robotics](#)
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.



- **Video Demonstration:**

[RootWave Demonstration1](#)

[RootWave Demonstration2](#)

2.2. Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO₂ lasers for precision weed eradication.
- **Limitations:** Not designed specifically for tall or toxic plants like hogweed; more suitable for row crops.





- **Video Demonstration:** [LaserWeeder Overview](#)

2.3. Odd.Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Suitability:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.



2.4. Drone and Aerial Detection Platforms (DK, DE, UK)

- **Application:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Complementary Role:** Ideal for pre-mapping and targeting robotic ground intervention. 



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2.5 [FarmWise Titan FT-35](#) (USA)

FarmWise (manufacturer)

Technology: self-driving robotic cultivator with multi-camera computer vision and trained AI and mechanical blades. (The system can also spot-apply chemicals like fertilizer or pesticides if needed.)

2.6 [Naïo Technologies “Dino”](#) (France)

Naïo Technologies (manufacturer)

Technology: **GPS guidance, lasers, and cameras** to localize itself and follow crop rows. Dino does **mechanical weeding** with adjustable harrows and hoeing tools that uproot weeds between crop rows. It can optionally use machine vision (as on Naïo’s smaller Oz robot) to detect crop rows or plants and skip crop plants while cultivating around them.

2.7 [ecoRobotix AVO \(Solar Weeding Robot\)](#) (Switzerland)

ecoRobotix (manufacturer)

solar-powered autonomous rover that identifies weeds and spot-sprays them with micro-doses of herbicide. AVO uses **cameras and AI software** to differentiate weeds from crops at the ground level. When a weed is detected, a robotic arm with a small spray nozzle precisely applies a few drops of herbicide onto the weed, minimizing chemical usage. (In tests it initially recognized two crop types – sugar beets and canola – and treated weeds in those fields.)

2.8 [FarmDroid FD20](#) (Denmark)

FarmDroid (manufacturer)

it **records the GPS coordinates of each seed** it plants, then later weeds the field based on those known positions. It is a solar-powered autonomous planter/weeder that first sows crops in precise grid patterns. During weeding, it does **blind mechanical cultivation** both *between rows and within rows* – any plant not at a recorded crop position is assumed to be a weed and is cut by its small hoes. This eliminates weeds even very close to the crop seedlings, without needing cameras.

2.9 [Small Robot Company “Tom & Dick” \(UK\)](#)

Small Robot Company (startup)

This UK-based system uses **two robots** in sequence. **Tom** is a lightweight scouting robot that autonomously maps the field, using cameras to capture detailed images of crops and weeds. The data is processed by an AI engine (nicknamed “Wilma”) which uses **pattern recognition** to distinguish weed seedlings from crop plants. Then **Dick**, the killing robot, navigates to the weed locations. For removal, Dick uses an electrical weeding system: it precisely positions a **small electrode probe** above each weed and delivers an ~8,000-volt electric shock to “boil” the weed from the roots. The weed-zapping arm is mounted on an igus delta robotic arm for speed and accuracy, and multiple arms can operate in parallel.

2.10 [AutoWeed \(JCU\) – Lantana Targeting Robot \(Australia\)](#)

James Cook University research (Australia)

Originally developed as a PhD project, this prototype robot focuses on spotting and spraying **Lantana camara**, a highly invasive shrub in Australia. It carries **eight cameras** feeding an onboard image processing AI. The system uses color (green vs brown), leaf shape, and texture analysis to identify the target weed (lantana) among native vegetation. When it recognizes lantana, it uses a mounted sprayer to apply a targeted dose of herbicide directly onto that plant. The robot navigates autonomously (currently towed or escorted by an ATV during tests) and geo-tags the locations of sprayed weeds via GPS for record-keeping.

2.11 [BirdHabitatBot \(UConn\) – Forest Invasives Robot \(USA\)](#)

BirdHabitatBot (UConn startup project)

BirdHabitatBot is a prototype system aimed at **forest understory invasive shrubs**, specifically Japanese barberry and multiflora rose in Northeast U.S. forests. The approach uses **drones for detection** and a ground robot for removal. First, a drone surveys the forest in early spring (a brief window when the invasives leaf out before native plants). Using deep learning, the system analyzes aerial images to **identify patches of the target invasive plants** by their spectral and textural signatures in the understory. These locations are mapped (GPS coordinates). Then a mobile robot (“BirdBot”) navigates to those coordinates on the forest floor to physically **cut the invasive plants** at the base. The removal robot is being designed to safely traverse leaf litter, rocks, and saplings to reach each plant and cut or pull it out.

2.12 [Bosch Deepfield “BoniRob” \(Germany\)](#)

Bosch (Farming – research prototype)

Uses Camera vision distinguishes weed seedlings by leaf shape with A piston punches weeds into soil (“stamping” method). No chemicals; very fast actuation (up to 1.75 weeds/sec). Prototype not commercialized; requires small weed size; heavy platform with generator

2.13 [“Hydrilla Hunter” Robotic Boat \(USA\)](#)

Northeastern University & Connecticut Ag. Experiment Station

Aquatic – freshwater weeds with Hyperspectral camera detects unique spectral signature

Comparison Table of Weed Removal Robots

Robot (Origin)	Environment & Use	Recognition & Mapping	Removal Mechanism 	Key Pros	Key Cons
Carbon LaserWeeder G2 (USA)	Farming – row crops (vegetables, row herbs)	Cameras + deep-learning AI distinguish crops vs weeds	8× high-powered lasers “zap” weeds (no contact)	No  herbicides; no soil disturbance; fast, day/night	Expensive; heavy tractor mount; limited to spaced row crops
FarmWise Titan FT-35 (USA)	Farming – vegetable fields	Multi-cameras + AI model trained on millions of images	Automated blades cut weeds around crop plants	Chemical-free weeding; adaptable to many crops; as-a-service model	Large machine; requires row structure; service fees (not owned)
Naïo Dino (France)	Farming – vegetable bed weeding	GPS guidance + front laser/camera for row positioning	Mechanical cultivator blades/hoes uproot weeds	Battery-electric (6–8h); reduces hand labor; multi-task capable	Heavy unit; high upfront cost; mainly inter-row (in-row weeds harder)

ecoRobotix AVO (Switzerland)	Farming – precision spot-spraying	Camera vision + AI to detect weeds among crops	Micro-dose herbicide sprayer on robotic arm	Solar-powered; uses 95% less herbicide; light & low compactio n	Slow coverage; limited initial crop/weed range; not for tall crops
FarmDroid FD20 (Denmark) 	Farming – seeding & weeding	RTK GPS maps crop seed positions (no vision needed)	Tiny harrows weed <i>between</i> and <i>within</i> rows mechanically	No camera needed (simple); zero chemicals; 24/7 solar run	Must use robot for planting; slow but steady; limited to row crops
Small Robot Co. Tom/Dick (UK) 	Farming – arable crops (trials)	“Tom” maps field with cameras; AI (Wilma) identifies weeds	“Dick” robot zaps weeds with 8kV electrical probe	No herbicides; can selectively leave benign plants; little soil impact	Prototype stage; two-step process is time-consuming; complex multi-robot system
AutoWeed Lantana Robot (Australia) 	Rangeland/Forestry – invasive shrubs	8 RGB cameras + AI vision (color/shape/text ure) find target weed	Sprays herbicide on detected weed foliage via nozzle	Cuts labor & cost for weed crews; 90% ID accuracy in tests	Prototype (tethered in tests); uses chemicals; terrain navigation challenges

BirdHabitat Bot (USA) 	 Forestry – understory invasives	Drone aerial imagery + deep learning to map invasive shrubs	Ground robot drives to locations and cuts shrubs at base	Targets only invasives (protects natives); keeps workers safe	Experimental ; rough terrain obstacles; may need supplementary herbicide for stumps
Hydrilla Hunter (USA) 	 Aquatic – freshwater weeds	Hyperspectral camera detects unique spectral signature	Marks GPS coordinates of hydrilla patches (for later removal)	Automates weed surveys; very species-specific (few false alarms)	Doesn't remove weed itself; limited battery; one-species focus (Hydrilla only so far)
Bosch Deepfield “BoniRob”  (Germany)	 Farming – research prototype	Camera vision distinguishes weed seedlings by leaf shape	A piston punches weeds into soil (“stamping” method)	No chemicals; very fast actuation (up to 1.75 weeds/sec)	Prototype not commercialized; requires small weed size; heavy platform with generator

3. Patent Research Overview

3.1. EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in giant hogweed without full severing. Minimizes regrowth by damaging vascular tissues.
- **Significance:** Basis for potential robotic adaptation using automatic puncturing probes.
- **Link:** [Patent EP3804483A1](#)

3.2. WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon Robotics[Dreamstime+3Log in or sign up to view+3GeekWire+3](#)
- **Description:** Robotic platform with sensors and vision system guiding a laser source to ablate unwanted plants.
- **Limitation:** Not optimized for toxic plants or non-row field applications.
- **Link:** [WO2021055485A1](#)

3.3. US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using capacitive and optical sensors for precision removal using mechanical tools.
- **Potential Use:** Adaptable platform for invasive species differentiation and removal.
- **Link:** [US20170238460A1](#)

3.4. EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH (Germany)
- **Description:** Electrically driven weed inactivation device using contact electrodes for plant-specific destruction.
- **Relevance:** Can be integrated into autonomous platforms for field use against large invasive targets.
- **Link:** [EP3744173A1](<https://patents.google.com/patent/EP3744173A1/en>)

